

One variable in one file: a single declaration and multiple usage sites

The localization unit for a variable-naming refactoring spans every occurrence of the symbol.

● ● ● OrderService.java

```
1 public int computeTotal(List<Order> orders) {  
2     int s = 0;  
3     for (Order o : orders) {  
4         s += o.getAmount();  
5     }  
6     if (s > LIMIT) {  
7         s = applyCap(s);  
8     }  
9     return s;  
10 }
```

same symbol

VARIABLE

s

1 declaration site
where the variable is introduced

5 usage references
read / write across 4 lines

Every occurrence binds to the same symbol, so a correct refactoring must localize the declaration **and** all usage sites together.